2006年真题参考答案

一、填空题

(1)1. (2)$2e^3$. (3)$4dx - 2dy$. (4)2. (5)$\frac{1}{9}$. (6)2.

二、选择题

(7) A. (8) C. (9) D. (10) B. (11) D. (12) A. (13) B. (14) A.

三、解答题

(15)(Ⅰ)$\frac{1}{x} - \frac{1-\pi x}{\arctan x}$. (Ⅱ)$\pi$.

(16)$\frac{2}{9}$.

(17) 证明略.(可考虑$f(x) = x\sin x + 2\cos x + \pi x$, $x \in [0,\pi]$,利用导数分析$f(x)$的单调性.)

(18)(Ⅰ)$y = ax^2 - ax$. (Ⅱ)$a = 2$.

(19) 收敛域为$[-1,1]$;$S(x) = 2x^2\arctan x - x\ln(1+x^2)$, $x \in [-1,1]$.

(20) 当$a = 0$时,$\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$线性相关,$\boldsymbol{\alpha}_1$为$\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$的一个极大线性无关组,且$\boldsymbol{\alpha}_2 = 2\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_3 = 3\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_4 = 4\boldsymbol{\alpha}_1$.

当$a = -10$时,$\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$线性相关,$\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$为$\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_4$的一个极大线性无关组,且$\boldsymbol{\alpha}_1 = -\boldsymbol{\alpha}_2 - \boldsymbol{\alpha}_3 - \boldsymbol{\alpha}_4$.

(21)(Ⅰ)$\boldsymbol{A}$的特征值为0, 0, 3.属于特征值0的特征向量为$k_1\boldsymbol{\alpha}_1 + k_2\boldsymbol{\alpha}_2$,其中$k_1$, k_2为不全为零的任意常数,属于特征值3的特征向量为$k(1,1,1)^{\mathrm{T}}$,其中k为任意非零常数.

(Ⅱ)$\boldsymbol{Q} = \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}$, $\boldsymbol{\Lambda} = \begin{pmatrix} 0 & & \\ & 0 & \\ & & 3 \end{pmatrix}$.

(Ⅲ)$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, $\left(\boldsymbol{A} - \frac{3}{2}\boldsymbol{E}\right)^6 = \left(\frac{3}{2}\right)^6\boldsymbol{E}$.

(22)(Ⅰ)$f_Y(y) = \begin{cases} \frac{3}{8\sqrt{y}}, & 0 < y < 1, \\ \frac{1}{8\sqrt{y}}, & 1 \leqslant y < 4, \\ 0, & \text{其他}. \end{cases}$ (Ⅱ)$\mathrm{Cov}(X,Y) = \frac{2}{3}$. (Ⅲ)$\frac{1}{4}$.

(23)(Ⅰ)$\hat{\theta} = \frac{3}{2} - \overline{X}$. (Ⅱ)$\hat{\theta} = \frac{N}{n}$.